

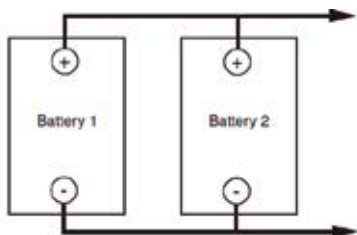
together, and there are two different approaches to do that:

To increase voltage, connect the batteries in series. The resultant voltage is the sum of the voltage outputs of all the cells combined.

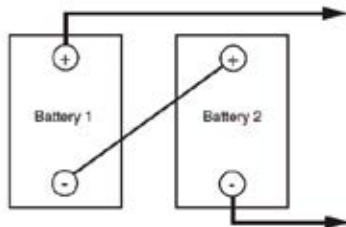
To increase current, connect the batteries in parallel. The resultant current is the sum of the current capacities of all the cells combined.

A point to be noted here that when you connect cells together not all cells may be discharged or recharged at the same rate, like the case when you have some new batteries connected with some half used batteries. The new batteries would provide most of the power and would get discharged earlier than the half used batteries.

Therefore, you should always replace or recharge all the cells together.



Parallel Connection (Current becomes twice)



Series connection (Voltage becomes twice)

Also, you can use pencil cells in series to get 9V or 12V (whatever voltage you require). Generally, batteries form the heaviest part of the robot, and so it is always advisable to keep the batteries as low as possible. Doing this would lower down the centre of mass of your robot and will make it more stable and less prone to toppling.

Taking care of your batteries and maintaining them for long time is very important. Follow the instructions given below to maintain your batteries strength for a long time:

Always keep your batteries dry. Do not short the battery terminals.

Remove battery from the holder when they are not in use. In case of NiCd batteries, discharge them fully before charging again, this would avoid development of memory effect in them. Do not expose batteries to high temperatures if you use them in a robot, keep the electronics devices which get heated very fast (like voltage regulators) away from them. Fix your battery to the base of your robot with Velcro, since Velcro has a good hold to keep batteries in place, and you can also remove the batteries easily if you want to replace or recharge them.